

Total No. of Questions : 8]

SEAT No. :

PB-4749

[Total No. of Pages : 2

[6263]-421

B.E. (Mechanical / Automobile)

**HONORS IN MODELLING AND SIMULATION OF EHV
(2019 Pattern) (Semester - VII) (302034MJ)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Figures to the right indicate full marks.
- 3) Draw the neat sketch wherever necessary.

- Q1)** a) Explain A Motor Control Unit in electric vehicle with its functional features. **[4]**
- b) Explain the following concepts related to modes of BMS control. **[8]**
- i) Centralized BMS Architecture
 - ii) Modular BMS Topology
 - iii) Primary/Subordinate BMS
 - iv) Distributed BMS Architecture
- c) What is Battery/Cell Control System in EV? Explain with its functions and working principal? **[8]**

OR

- Q2)** a) Explain Torque and speed coupling in electric vehicle. **[4]**
- b) Explain following motor Control methods (any four). **[8]**
- i) Phase flux linkage
 - ii) Phase Inductance
 - iii) Modulated Signal Injection
 - iv) Mutually Induced Voltage
 - v) Observer based Theory
- c) Explain the importance of Sensor Management and Integration for electric vehicle with functional and application point of view. **[8]**

P.T.O.

- Q3)** a) Explain, “Placement of Motors” in electric vehicle system. [8]
b) Explain the important feature of following (any four) : [8]
i) Unicycle
ii) Bicycle
iii) Dicycle
iv) Tricycle
v) Quadracycle

OR

- Q4)** a) Explain “Batter and Motion Transmission Systems” in electric vehicle system. [8]
b) Explain Propulsion and Power distribution system in electric vehicle with its main components. [8]

- Q5)** a) Explain Design and repair tips for frame of vehicles. [8]
b) Explain the types and components of battery pack in electric vehicle. [8]

OR

- Q6)** a) Explain the types of chassis on basis of following : [8]
i) Engine Location
ii) Assembly
iii) Frame
b) Explain the following parameters related to vehicle dynamics in detail. [8]
i) Aerodynamic drag
ii) Aerodynamic lift
iii) Side force
iv) Rolling movement

- Q7)** a) List out the feature of Electric Vehicle Dashboard considering suitable example of 2W/3W/4W. [10]
b) What do you mean by ergonomics and aesthetics aspects of ev? Explain in detail with important features? [8]

OR

- Q8)** a) Explain Noise Factors & Failure Modes in electric vehicle. [10]
b) Explain the important features of FMEA analysis in electric vehicle. [8]

